

CCNA 200-301 Quick Reference

1 Layer 1 Standards

1.1 ANSI/TIA T568 Pinout Standards

Pin	T568A	T568B	MDI	MDI-X
1	Green-White	Orange-White	Tx	Rx
2	Green	Orange	Tx	Rx
3	Orange-White	Green-White	Rx	Tx
4	Blue	Blue		
5	Blue-White	Blue-White		
6	Orange	Green	Rx	Tx
7	Brown-White	Brown-White		
8	Brown	Brown		

Two standard T568 pinouts exist for the *Medium-Dependent Interface* (MDI) and *MDI Crossover* (MDI-X) port standards. Computers, routers, and Layer 3 devices use the MDI standard; hubs and Layer 2 devices generally use the MDI-X standard. Connections between MDI and MDI-X ports use a *straight-through* cable, where the same T568 standard terminates each end of the cable. Connections between two MDI (or two MDI-X) ports use a *crossover* cable, where different T568 standards terminate each end of the cable. Most new equipment incorporates *Auto-MDIX* circuitry to allow either cable type.

Note: The T568B standard is more commonly used for straight-through cables in North America.

1.2 Cabling Standards

Table 1.1: IEEE 802.3 Copper Standards

802.3i	10BASE-T	10 Mbps	CAT3
802.3u	100BASE-TX	100 Mbps	CAT5
802.3ab	1000BASE-T	1 Gbps	CAT5e
802.3z	1000BASE-SX 1000BASE-LX	1 Gbps 550m Fiber 5km Fiber	
802.3an	10GBASE-T	10 Gbps	CAT6a
802.3ba	40GBASE-T	40 Gbps	CAT8
802.3bz	2.5GBASE-T 5GBASE-T	2.5 Gbps 5 Gbps	CAT5e CAT5e

Table 1.2: SFP Transceivers

SX	MMF 850 nm	550 m
SR	black	300 m
LX/LR	SMF 1310 nm blue	10-20 km
—	SMF 1490 nm violet	—
EX/ER ZX/ZR	SMF 1550 nm yellow	40 km 80 km

Table 1.3: ANSI/TIA-598 Fiber Standards

Type	Polish	Jacket/Connector	Usage
OS1/OS2	UPC	yellow/blue	indoor/outdoor
	APC	yellow/green	
OM1	UPC	orange/beige orange/gray	100 Mbps
OM2	UPC	orange/black	1 Gbps
OM3/OM4	UPC	aqua/aqua	10/40/100 Gbps
OM5	UPC	lime/lime	10/40/100 Gbps

Table 1.4: Power over Ethernet (PoE) Standards

Cisco Inline Power	ILP	7.5W	48v	Mode B
IEEE 802.3af	Type 1 PoE	class 0-3 15.4W	44-57v	≥ CAT3 Mode A/B
IEEE 802.3at	Type 2 PoE+	class 0-4 30W	50-57v	≥ CAT5 Mode A/B
IEEE 802.3bt	Type 3 UPoE	class 0-6 60W	50-57v	≥ CAT5 4Pair PoE
	Type 4 UPoE+	class 0-8 100W	52-57v	

IEEE *Active PoE* negotiates power delivery via LLDP. Nonstandard *Passive PoE* provides a constant voltage that may **damage** equipment. **Mode A** PoE energizes the 2 data wire pairs; **Mode B** PoE energizes the 2 spare wire pairs; **4Pair** PoE energizes all 4 wire pairs.

2 Layer 2 Protocols

Table 2.1: EtherType Values

0x0800	IPv4 Packet
0x0806	ARP
0x2000	Cisco CDP
0x8100	IEEE 802.1Q
0x86dd	IPv6 Packet
0x8809	Ethernet Slow Protocols
0x8847	MPLS Unicast
0x8848	MPLS Multicast
0x8863	PPPoE Discovery
0x8864	PPPoE Session
0x888e	IEEE 802.1x EAPOL
0x88cc	IEEE 802.1AB LLDP

Table 2.2: Layer 2 Multicast

0000.0c07.acXX	Cisco HSRPv1
0000.0c9f.fXXX	Cisco HSRPv2
0000.5e00.01XX	RFC 5798 VRRP
0007.b40X.XXY	Cisco GLBP
0100.5eXX.XXXX	IPv4 Multicast
3333.XXXX.XXXX	IPv6 Multicast
0100.0ccc.cccc	Cisco PAgP / CDP / VTP
0100.0ccc.cccd	Cisco PVST+ / PVRST
0180.c200.0000	IEEE 802.1D STP
	IEEE 802.1w RSTP
0180.c200.0002	IEEE 802.3ad LACP
0180.c200.000e	IEEE 802.1AB LLDP

Note: 0180.c200.000X multicast addresses are not forwarded by 802.1D-compliant bridges. Not all Layer 2 protocols support multicast.

2.1 Ethernet II / IEEE 802.3

Preamble	7 Bytes	10101010 (0xaa)
SFD	1 Byte	10101011 (0xab)
Destination MAC	6 Bytes	
Source MAC	6 Bytes	
Type (Ethernet II) Length (802.3)	2 Bytes	EtherType (Table 2.1) Payload Length in Bytes
Data	varies	46-1500 Bytes based on MTU
FCS	4 Bytes	

Table 2.3: IEEE 802.2 Logical Link Control (LLC)

Destination Service Access Point (DSAP)	1 Byte
Source Service Access Point (SSAP)	1 Byte
Control	1-2 Bytes

Table 2.4: IEEE SubNetwork Access Protocol (SNAP) Header

Organizationally Unique ID (OUI)	3 Bytes	0x000000 or Vendor OUI
Type	2 Bytes	EtherType (Table 2.1)

Note: Each 802.3 frame includes an 802.2 LLC header for process multiplexing. Per RFC 1042, an 802.2 LLC header with value 0xaaaa03 includes a SNAP header identifying the EtherType.

2.2 VLAN Tagging

Table 2.5: IEEE 802.1Q Tagging

Tag Protocol Identifier (TPID)	16 bits	EtherType (0x8100, Table 2.1)
TCI: Priority Code Point (PCP)	3 bits	QoS Class-of-Service (CoS) Marking (Section 6)
TCI: Drop Eligible Indicator (DEI)	1 bit	QoS Drop Eligibility
VLAN ID	12 bits	

Table 2.6: Cisco InterSwitch Link (ISL)

Destination Address	40 bits	Multicast 0100.0c00.00 or 0300.0c00.00
Type	4 bits	Frame Type
User	4 bits	Priority Handling
Source Address	48 bits	Sending Switchport MAC
LEN	16 bits	Original Frame Length in Bytes
AAAA03	24 bits	SNAP/LLC Field (0xaaaa03)
HSA	24 bits	Source Address OUI (0000.0c)
VLAN	15 bits	VLAN ID
BPDU Flag	1 bit	Flag BPDU / CDP / VTP frames
INDEX	16 bits	Diagnostics
RES	16 bits	Token Ring / FDDI Frames
Data	varies	8-196000 bit Unmodified original frame
FCS	32 bits	

Note: Industry-standard IEEE 802.1Q supports both *normal-range* VLANs (1-1005) and *extended-range* VLANs (1006-4094); Cisco ISL is deprecated.

2.3 Cisco VLAN Trunking Protocol (VTP)

Cisco VTP dynamically advertises a *VLAN Database* to participating switches over Cisco ISL and IEEE 802.1Q trunk links. Participating switches use either *VTP Server Mode* or *VTP Client Mode*, and default to VTP Server Mode with a **Config Revision Number** of 0. Each time a VTP Server's VLAN Database is locally updated, the **Config Revision Number** increments and *VTP Synchronization* occurs.

VTP Synchronization uses a *VTP Summary Advertisement* (Table 2.7) alongside 1 or more *VTP Subset Advertisements* (Table 2.8) to advertise the revised VLAN Database. VTP Servers also send out a Summary Advertisement (alongside 0 or more Subset Advertisements) every 5 minutes. To participate, newly-connected switches may send a *VTP Advertisement Request* (Table 2.10) over each trunk link that comes up. A switch only listens to VTP messages in its **VTP Management Domain** (default NONE). An MD5-hashed **VTP Password** (default NONE) can be configured to prevent unauthorized switches from participating in the VTP Management Domain.

VTP Clients participate in VTP Synchronization but disallow local VLAN configurations. *VTP Transparent Mode* or *VTP Off Mode* switches use a local VLAN configuration instead of the VLAN Database, although they may forward VTP messages.

Table 2.7: VTP Summary Advertisement

Version	1 Byte	VTP Version (1-3)
Code	1 Byte	Summary Advertisement (0x01)
Followers	1 Byte	Indicates a Subset Advertisement
MgmtD Len	1 Byte	
Management Domain	32 Bytes	VTP Domain Name
Config Revision Number	4 Bytes	
Updater Identity	4 Bytes	Originating VTP Server (IP Address)
Update Timestamp	12 Bytes	Datetime of revision
MD5 Digest	16 Bytes	VTP Password hash (if configured)

Table 2.8: VTP Subset Advertisement

Version	1 Byte	VTP Version (1-3)
Code	1 Byte	Subset Advertisement (0x02)
Sequence Number	1 Byte	Sequence in packet stream following Summary Advertisement
MgmtD Len	1 Byte	
Management Domain	32 Bytes	VTP Domain Name
Config Revision Number	4 Bytes	
VLAN-Info Field(s)	4 Bytes	Advertised VLAN(s) (Table 2.9)

Table 2.9: The VLAN-Info Field

V-Info-Len	1 Byte
Status	1 Byte
VLAN-Type	1 Byte
VLAN-Name Len	1 Byte
ISL VLAN-ID	2 Bytes
MTU Size	2 Bytes
802.10 Index	4 Bytes
VLAN-Name	4 Bytes

Table 2.10: Advertisement Requests

Version	1 Byte	VTP Version (1-3)
Code	1 Byte	Advertisement Request (0x03)
Reserved	1 Byte	
MgmtD Len	1 Byte	
Management Domain	32 Bytes	VTP Domain Name
Start-Value	32 Bytes	Identifies the requested Subset Advertisement

2.4 Cisco Discovery Protocol (CDP) / IEEE 802.1AB Link-Layer Discovery Protocol (LLDP)

Table 2.11: CDP Frame Format

Version	1 Byte	CDP Version (1-2)
Time-to-Live (TTL)	1 Byte	CDP Hold timer
Checksum	2 Bytes	
TLV List	varies	CDP TLV(s)

Table 2.12: CDP TLVs

Type	2 Bytes
Length	2 Bytes
Value	varies

CDPv2 advertises Cisco device information to directly-connected neighbors. Each device maintains a *CDP Neighbor Table* based on received CDP messages. It is necessary for Cisco IP Telephony and some other features.

Table 2.13: IEEE 802.1AB LLDP PDU Format

Chassis ID TLV	9 Bytes	Source MAC (Type 1)
Port ID TLV	6 Bytes	Source Interface (Type 2)
Time-to-Live TLV	4 Bytes	LLDP Hold timer (Type 3)
Optional TLVs	varies	Optional LLDP TLV(s) (Types 4-127)
End TLV	2 Bytes	End of LLDP PDU (Type 0)

Table 2.14: LLDP TLVs

Type	7 bits
Length	9 bits
Value	0-511 Bytes

LLDP advertises vendor-neutral device information to directly-connected neighbors. Each device maintains a *LLDP Neighbor Table* based on received LLDP messages. LLDP devices can autonegotiate the use of ANSI/TIA-1057 *LLDP Media Endpoint Discovery* (LLDP-MED), providing advanced autodiscovery, VoIP, PoE, and other features.

2.5 IEEE 802.1D Spanning-Tree Protocol (STP) / IEEE 802.1w Rapid STP (RSTP)

Protocol Identifier	2 Bytes	STP / PVST+ (0x0000)
Version	1 Byte	STP / PVST+ (0x00) RSTP / PVRST (0x02)
Message Type	1 Byte	Identify Configuration / TCN BPDUs RSTP/MSTP (0x02)
Flags	1 Byte	Signals TC / TCA bits
Root ID	8 Bytes	The sender's Root BID (Table 2.15)
Root Path Cost	4 Bytes	The sender's cost to Root
Bridge ID	8 Bytes	The sender's BID (Table 2.15)
Port ID	2 Bytes	The sender's Port Prio.Nbr
Message Age	2 Bytes	Time since Root sent this BPDU
Max Age	2 Bytes	Time until BPDU expires
Hello Time	2 Bytes	How often Root sends BPDUs
Forward Delay	2 Bytes	Time spent in each transition state

Table 2.15: Spanning Tree Bridge ID (BID) Format

Base Priority	4 bits	Configured bridge priority (multiple of 4096)
System ID Extension	12 bits	The VLAN ID of this STP instance
System ID	48 bits	The bridge <i>Burned-In Address</i> (BIA)

Table 2.16: Spanning Tree Port Costs

Port Speed	STP IEEE Cost	Revised IEEE Cost	RSTP IEEE Cost
10 Mbps	100	100	2,000,000
100 Mbps	10	19	200,000
1 Gbps	1	4	20,000
10 Gbps	1	2	2,000
100 Gbps	-	1	200
1 Tbps	-	-	20
10 Tbps	-	-	2

Table 2.17: Spanning Tree Port States

STP State	RSTP State	Send/Receive BPDUs	Forward Data	Learn MACs
Disabled	Discarding	No	No	No
Blocking	Discarding	Receive	No	No
Listening	—	Yes	No	No
Learning	Learning	Yes	No	Yes
Forwarding	Forwarding	Yes	Yes	Yes

The *STP Convergence Process*:

1. Elect Root Bridge:

- (a) Lowest received BPDUs Root ID: Base Priority
- (b) Tiebreaker: lowest received BPDUs Root ID: System ID

2. Elect *Root Ports* (RPs):

- (a) Lowest received BPDUs Root Path Cost + local Port Cost (Table 2.16)
- (b) Tiebreaker: lowest received BPDUs BID
- (c) Tiebreaker: lowest received BPDUs Port ID: Prio.Nbr
- (d) Other RSTP link-type point-to-point ports become *Alternate Ports* (APs)

3. Elect *Designated Ports* (DPs):

- (a) Lowest advertised BPDUs Root Path Cost
- (b) Tiebreaker: lowest advertised BPDUs BID
- (c) Tiebreaker: lowest advertised BPDUs Port ID: Prio.Nbr
- (d) Other RSTP link-type shared ports on the same bridge become *Backup Ports* (BPs)

4. Other Ports:

- (a) Working ports become STP *Nondesignated Ports* (NDs)
- (b) Nonworking and disabled ports become STP disabled ports

Optional STP Features:

PortFast: Locally or globally configured ports immediately enter Forwarding state on edge ports. BPDUs are sent, but received BPDUs disable the feature.

BPDUs Guard: Locally or globally configured (PortFast) ports prevent unauthorized devices from altering the STP topology. BPDUs are sent, but received BPDUs err-disable the port. Compatible with globally-configured BPDUs Filter.

BPDUs Filter: Locally configured ports ignore STP; globally configured (PortFast) ports do not send BPDUs. Compatible with BPDUs Guard.

Root Guard: Locally configured ports protect the STP Root Bridge. BPDUs are sent, but received superior BPDUs err-disable the port (Root Inconsistent state). Incompatible with Loop Guard.

Loop Guard: Locally or globally configured (point-to-point ports) protect against unidirectional links, preventing RPs and NDs from becoming DPs. Ports whose Max Age reaches 0 err-disable (Loop Inconsistent state). Incompatible with Root Guard.

2.6 RFC 826 Address Resolution Protocol (ARP)

HTYPE	16 bits	L2 Protocol (1 for Ethernet)
PTYPE	16 bits	L3 Protocol (EtherType, Table 2.1)
HLEN	8 bits	L2 Address Length in Bytes
PLEN	8 bits	L3 Address Length in Bytes
Operation	16 bits	Message Type (1 for Request, 2 for Reply)
Origin HW	48 bits	Source L2 Address
Origin IP	32 bits	Source L3 Address
Target HW	48 bits	Destination L2 Address
Target IP	32 bits	Destination L3 Address

Note: As a Layer 2 protocol, all ARP messages are encapsulated directly within an Ethernet frame.

Dynamic ARP Inspection (DAI) is an optional switch security feature to prevent ARP Poisoning and ARP DoS. The *DHCP Snooping Binding Table* is used to filter incoming ARP messages based on their **Origin HW** and **Origin IP** values; an ARP ACL can also be configured for hosts using static IP addresses. Optional verification checks compare the ARP **Origin/Target HW** values against the frame **Source/Destination MAC** and ensure the ARP **Origin/Target IP** fields contain unicast values. This behavior is disabled on DAI *trusted ports*. DAI uses optional per-interface rate limits to prevent DoS attacks against the switch CPU and ARP table.

2.7 IEEE 802.11 Wireless LANs (WLANs)

Frame Control	2 Bytes	Message Type / Subtype
Duration / ID	2 Bytes	Frame Transmission Time / Client Association
Address 1	6 Bytes	Dst / Src / Rx / Tx Address
Address 2	6 Bytes	Dst / Src / Rx / Tx Address
Address 3	6 Bytes	Dst / Src / Rx / Tx Address
Sequence Control	2 Bytes	Fragmentation / Duplication Management
Address 4	6 Bytes	Dst / Src / Rx / Tx Address
QoS Control	2 Bytes	Section 6
HT Control	4 Bytes	HT Operations (802.11n and later)
Data	varies	
FCS	4 Bytes	

In America, the 2.4-GHz ISM band has 11 overlapping channels (1/6/11 nonoverlapping) and the 5-GHz U-NII band has 23 nonoverlapping channels. WLANs operate at half duplex with *Carrier Sense Multiple Access / Collision Avoidance* (CSMA/CA) to minimize collisions.

The original IEEE 802.11-1997 standard uses *Frequency Hopping Spread Spectrum* (FHSS) encoding in the 2.4-GHz band without channels; each consecutive transmission uses a slightly different frequency to minimize collision risk. Modern IEEE 802.11 standards use either *Direct Sequence Spread Spectrum* (DSSS) or *Orthogonal Frequency Division Multiplexing* (OFDM) encoding. DSSS uses the 2.4-GHz band with a 22 MHz channel width; OFDM uses either band with a 20 MHz channel width.

WLANs use a *Wi-Fi Protected Access* (WPA) standard to ensure message privacy and integrity for wireless-frames in-transit. Devices authenticate with the AP during the association process. Clients authenticate using either a *Pre-Shared Key* (PSK) or IEEE 802.1x Access Control with an IEEE 802.11i *Extensible Authentication Protocol* (EAP) framework. See Table 8.7 for details.

Table 2.18: IEEE 802.11 Standards

Standard	RF Band	Bandwidth	Encoding	Name
-1997	2.4 GHz	2 Mbps	FHSS/DSSS	
b	2.4 GHz	11 Mbps	DSSS	
a	5 GHz	54 Mbps	OFDM	
g	2.4 GHz	54 Mbps	OFDM	
n	2.4 / 5 GHz	600 Mbps	OFDM	Wi-Fi 4 (HT)
ac	5 GHz	6.93 Gbps	OFDM	Wi-Fi 5 (VHT)
ax	2.4 / 5 GHz 6 GHz	9.6 Gbps	OFDM	Wi-Fi 6 (HE) Wi-Fi 6e
be	2.4/5/6 GHz	46 Gbps	OFDM	Wi-Fi 7 (EHT)

Table 2.19: WLC Deployment

Deployment	WLC Location	Clients	APs
Unified	Central	64,000	6,000
Cloud	Data Center	32,000	3,000
Embedded	Access Switch	4,000	200
Mobility Express	LAP	2,000	100

Table 2.20: WLC Ports and Interfaces

WLC Port	WLC Interface	VLAN	Usage
Console			Initial config
Service Port	Service Port	OOB-MGMT	Out-of-band management (access port) <i>bootup / system recovery</i>
Redundancy	Redundancy Management	MGMT	In-band management (standby WLC) <i>HA redundancy</i>
DS	Management	MGMT	In-band management (active WLC) <i>Form CAPWAP tunnels</i>
DS	Virtual	Mobility Group	DHCP Relay, WebAuth
DS	Dynamic	USERS	Bind WLANs to VLANs (tunnel LAG)

Note: Lightweight APs require a *Wireless LAN Controller* (WLC) and CAPWAP to support WLANs. Each Autonomous AP can independently support multiple WLANs.

Autonomous AP Modes:

Infrastructure: Offer BSS' on an RF Channel

Repeater: Extend a BSA via retransmission

WorkGroup Bridge (WGB): Bridge wired device(s) to a WLAN

Bridge: Form a Point-to-Point (P2P) / Point-to-Multipoint (P2MP) link between LANs

Mesh: Bridge traffic across APs in a large service area

Lightweight AP (LAP) Modes:

Local: Offer BSS' on an RF Channel

Monitor: Monitor for IDS events / rogue APs, determine STA positions

FlexConnect: Locally switch traffic if CAPWAP fails

Rogue Detector: Detect rogue devices (correlate wired and wireless MACs)

Sniffer: Capture WLAN traffic for analysis

Bridge: Form a Point-to-Point (P2P) / Point-to-Multipoint (P2MP) link or a mesh

Flex+Bridge: FlexConnect on a mesh LAP

SE-Connect: Detect interference sources (RF spectrum analysis)

2.8 ITU High-Level Data-Link Control (HDLC)

Flag	1 Byte	Synchronization
Address	1 Byte	Destination Node
Control	1 Byte	
Data	varies	
FCS	4 Bytes	

2.9 IETF RFC 1661 Point-to-Point Protocol (PPP) / Cisco HDLC (cHDLC)

Flag	1 Byte	Synchronization
Address	1 Byte	Destination Node
Control	1 Byte	
Type	2 Bytes	EtherType (Table 2.1)
Data	varies	
FCS	4 Bytes	

3 Layer 3 Protocols

Table 3.1: IPv4 Protocol / IPv6 Next Header Values

0x01	1	ICMP
0x02	2	IGMP
0x04	4	IPv4
0x06	6	TCP
0x11	17	UDP
0x29	41	IPv6
0x2f	47	GRE
0x3a	58	ICMPv6
0x58	88	EIGRP
0x59	89	OSPF
0x67	103	PIM
0x70	112	VRRP
0x89	137	MPLS-in-IP

Table 3.2: Layer 3 Multicast

224.0.0.1 ff02::1	All-IPv4-Nodes All-IPv6-Nodes	
224.0.0.2 ff02::2	All-IPv4-Routers All-IPv6-Routers	Cisco HSRPv1
224.0.0.5 ff02::5	All-SPF-Routers	OSPFv2 OSPFv3
224.0.0.6 ff02::6	All-SPF-DRs	OSPFv2 OSPFv3
224.0.0.9 ff02::9	All-RIP-Routers	RIPv2 RIPng
224.0.0.10 ff02::a	All-EIGRP-Routers All-EIGRPv6-Routers	EIGRP EIGRPv6
224.0.0.18	RFC 5798 VRRP	
224.0.0.102	Cisco HSRPv2 / GLBP	

Note: Each Layer 3 multicast address maps to a corresponding Layer 2 multicast address which may be used in the L2PDU.

The *Routing Information Base* (RIB, routing table) is populated based on each route's *Administrative Distance* (AD). Route metric is calculated by the routing protocol and acts as a tiebreaker for multiple routes to the same destination via the same routing protocol. Some routing protocols are capable of multi-path load-balancing, resulting in multiple valid routes to the same destination under specific conditions.

Table 3.3: Route Types

Route Type	AD Value	IGP Type
Connected	0	
Static	1	
BGP	20	EGP
EIGRP	90	Advanced Distance Vector
IGRP	100	Distance Vector
OSPF	110	Link-State
IS-IS	115	Link-State
RIP	120	Distance Vector
EIGRP External	170	Advanced Distance Vector
iBGP	200	EGP
DHCP	254	
Invalid	255	

3.1 RFC 791 IP Version 4

Version	4 bits	IP Version (4)
IP Header Length (IHL)	4 bits	Header Length (Bytes $\div$ 5)
DS Field	8 bits	QoS Type-of-Service (ToS) Marking (Section 6)
Packet Length	16 bits	Total Packet Length
Identification	16 bits	Fragmentation
Flags	3 bits	Fragmentation
Fragment Offset	13 bits	Fragmentation
Time-to-Live (TTL)	8 bits	Loop Prevention
Protocol	8 bits	Protocol Type (Table 3.1)
Header Checksum	16 bits	
Source IP	32 bits	
Destination IP	32 bits	
Options	varies	Optional Header Fields
Data	varies	

Table 3.4: RFC 791 / RFC 1918 Addressing

RFC 791	First Octet	Address Block	RFC 1918 Block
Class A	0XXX XXXX	0.0.0.0 - 127.0.0.0 /8	10.0.0.0 /8
Class B	10XX XXXX	128.0.0.0 - 191.255.0.0 /16	172.16.0.0 /12
Class C	110X XXXX	192.0.0.0 - 223.255.255.0 /24	192.168.0.0 /16
Class D	1110 XXXX	224.0.0.0 - 239.255.255.255	
Class E	1111 XXXX	240.0.0.0 - 255.255.255.255	

Table 3.5: IPv4 Subnetting Magic Numbers

Bit Position	1	2	3	4	5	6	7	8
Octet Mask	128	192	224	240	248	252	254	255
Addresses	128	64	32	16	8	4	2	1
Octet Wildcard	127	63	31	15	7	3	1	0

Note: RFC 3927 reserves block 169.254.0.0/16 for IPv4 Link-Local Addressing.

IPv4 hosts use an *ARP Table* (mapping Layer 2/3 addresses) and Layer 3 forwarding logic to send packets. Local traffic is forwarded directly; external traffic is forwarded to the default gateway address.

3.2 RFC 2460 IP Version 6

Version	4 bits	IP Version (6)
Traffic Class	8 bits	QoS Marking (Section 6)
Flow Label	20 bits	Experimental
Payload Length	16 bits	Data + Extension Headers Length
Next Header	8 bits	Protocol Type (Table 3.1)
Hop Limit	8 bits	Loop Prevention
Source Address	128 bits	
Destination Address	128 bits	
Data	varies	

Note: IPv6 uses a fixed 40-Byte header; IPv6 Options headers are sent separately.

Table 3.6: RFC 2460 Addressing

Address Class	Block	Address Format		
Global Unicast	Any	Prefix + P bits	Subnet ID + $64 - P$ bits	Interface ID + 64 bits
Unique Local Unicast	fd00::/8	fd + 8 bits	Global ID + 40 bits	Subnet ID + 16 bits Interface ID + 64 bits
Link-Local Unicast	fe80::/64		fe80:0:0:0 + 64 bits	Interface ID + 64 bits
Multicast	ff00::/8	ff02::1:ffXX:XXXX/104 Solicited-Node		
	ff01::/16	Interface/Node-Local		
	ff02::/16	Link-Local		
	ff05::/16	Site-Local		
	ff08::/16	Org-Local		
	ff0e::/16	Global		

Note: IPv6 does not support broadcasts, only scoped multicasts.

Modified EUI-64 Process for Unique Address Generation:

1. Split the interface MAC address into two 24-bit parts.
2. Insert 0xffffe between the two parts, creating a 64-bit Interface ID.
3. Invert the 7th bit (the *Universal/Local bit*) of the resulting Interface ID.

3.3 RFC 792 Internet Control Message Protocol (ICMP) / RFC 4443 ICMPv6

Type	8 bits	Message Type (Tables 3.7 and 3.8)
Code	8 bits	Message Subtype / Status Code (Tables 3.7 and 3.8)
Checksum	16 bits	
Header Data	32 bits	Message-specific fields
Payload	varies	

Table 3.7: ICMP Type.Code Values

0.0	Echo Reply
3.X	Destination Unreachable
8.0	Echo Request
9.0	Router Advertisement
10.0	Router Solicitation
11.X	Time Exceeded
30.0	Traceroute
42.0	Extended Echo Request
43.X	Extended Echo Reply

Table 3.8: ICMPv6 Type.Code Values

1.X	Destination Unreachable
2.0	Packet Too Big
3.X	Time Exceeded
128.0	Echo Request
129.0	Echo Reply
133.0	NDP RS
134.0	NDP RA
135.0	NDP NS
136.0	NDP NA
137.0	NDP Redirect

3.4 RFC 4861 ICMPv6 Neighbor Discovery Protocol (NDP)

ICMPv6 NDP offers core functionality for IPv6 networks, including *Neighbor Discovery*, *Router Discovery*, *Stateless Address Autoconfiguration* (SLAAC), and *Duplicate Address Detection* (DAD).

Router Solicitation (RS) and *Router Advertisement* (RA) messages (Tables 3.9 and 3.11) replace DHCP/DHCPv6 default gateway discovery. RS messages are sent to All-IPv6-Routers multicast; RA messages are sent either to a unicast target address or to All-IPv6-Hosts multicast.

Neighbor Solicitation (NS) and *Neighbor Advertisement* (NA) messages (Tables 3.10 and 3.12) replace ARP. NS messages are sent to a solicited-node multicast target address; NA messages are sent either to a unicast target address or to All-IPv6-Hosts multicast.

The SLAAC Process:

1. The IPv6 host learns the IPv6 prefix used on the link, from any router, using NDP RS/RA messages.
2. The IPv6 host builds an IPv6 unicast address using the learned prefix and a random/EUI-64 generated interface ID.
3. Before using the address, the IPv6 host uses DAD to ensure that no other IPv6 host is already using the same address.

The DAD Process:

1. The IPv6 host sends an NDP NS message, listing its own IPv6 unicast address as the **Target Address**.
2. If no other IPv6 host uses that address, then no host should reply with an NDP NA message; the host is safe to use that address on the IPv6 network.
3. If another IPv6 host uses that address, they reply with an NA message. The local host receives an NDP NA message and avoids using that address until the issue is resolved.

Table 3.9: NDP RS

Type	8 bits	Message Type (133)
Code	8 bits	Message Subtype (0)
Checksum	16 bits	
Reserved	32 bits	Unused (0)
Options	varies	

Table 3.10: NDP NS

Type	8 bits	Message Type (135)
Code	8 bits	Message Subtype (0)
Checksum	16 bits	
Reserved	32 bits	Unused (0)
Target Address	128 bits	
Options	varies	

Table 3.11: NDP RA

Type	8 bits	Message Type (134)
Code	8 bits	Message Subtype (0)
Checksum	16 bits	
Current Hop Limit	8 bits	Default IPv6 Hop Count Value
Managed Address Flag	1 bit	Indicates DHCPv6 Address Services
Other Config Flag	1 bit	Indicates DHCPv6 Other Services
Reserved	6 bits	Unused (0)
Router Lifetime	16 bits	Default Router Lifetime
Reachable Time	32 bits	Neighbor Unreachability Detection
Retransmit Time	32 bits	Time between NS message retransmissions (ms)
Options	varies	

Table 3.12: NDP NA

Type	8 bits	Message Type (136)
Code	8 bits	Message Subtype (0)
Checksum	16 bits	
From Router Flag	1 bit	Identifies sender as router
Solicited Flag	1 bit	Indicates solicitation by NDP NS
Override Flag	1 bit	Overrides existing cache entry
Reserved	29 bits	Unused (0)
Target Address	128 bits	
Options	varies	

Note: Unlike IPv4 ARP, ICMPv6 NDP messages are encapsulated within an IPv6 packet.

IPv6 hosts use an *NDP Neighbor Table* (mapping Layer 2/3 addresses) and an IPv6 routing table (populated by Router Discovery) to send packets. *On-link* traffic is forwarded directly; *off-link* traffic is forwarded to the default gateway's link-local address.

3.5 RFC 2328 Open Shortest Path First (OSPFv2)

OSPFv2 is a link-state IGP for dynamic routing which maintains a *Link-State Database* (LSDB) representing the internetwork of connected OSPF routers within an *OSPF area*. Each OSPF router forms OSPF neighborships with connected OSPF routers through a series of OSPF messages and *OSPF neighbor states*, which synchronize their LSDBs. Each OSPF area must connect to *backbone area 0* via an *OSPF Area Border Router* (ABR). An *OSPF Autonomous System Border Router* (ASBR) connects the local *Autonomous System* (AS) with one or more external AS.

Table 3.13: OSPF LSA Types

LSA Type	Source	Conditions / Notes
Type 1 Router	SPF-Routers	1 per intra-area router
Type 2 Network	SPF-DRs	1 per DR/BDR subnet
Type 3 Summary	ABRs	1 per inter-area subnet
Type 3 Default	ABRs	Advertises the local ABR
Type 4 Summary ASBR	ABRs	Advertises a non-local ASBR
Type 5 AS-External	ASBRs	1 per AS-External network
Type 7 NSSA	ASBRs	1 per AS-External network (stub area)

A *Stub Area* is a non-backbone OSPF area with Type 3 Default and Type 3 Summary LSAs. A *Totally Stubby Area* is a stub area with no Type 3 Summary LSAs. A *Not-So-Stubby Area* (NSSA) is a stub area with Type 7 LSAs. A *Totally NSSA* is a totally stubby area with Type 7 LSAs.

Table 3.14: OSPF Link-State Advertisement (LSA) Header

LS Age	2 Bytes	Seconds since LSA originated
Options	1 Byte	
LS Type	1 Byte	LSA Type (Table 3.13)
Link-State ID	4 Bytes	Determined by LS Type
Advertising Router	4 Bytes	Advertising Router RID
LS Sequence Number	4 Bytes	Used by routers to judge LSAs
LS Checksum	2 Bytes	Checksum of LSA (excluding LS Age)
Length	2 Bytes	LSA + Header Length in Bytes

Each LSDB is a collection of *OSPF Link-State Advertisements* (LSAs) whose contents describe the internetwork according to their LSA type (Table 3.13). Note that the number of OSPF routers, OSPF areas, and other variables determine the number and type of LSAs within the LSDB.

Table 3.15: OSPF Hello

Version	1 Byte	OSPF Version (2)
Type	1 Byte	OSPF Packet Type (1)
Packet Length	2 Bytes	Header + Packet Length in Bytes
Router ID	4 Bytes	Advertising Router RID
Area ID	4 Bytes	Advertising Router's Area ID
Checksum	2 Bytes	
AuType	2 Bytes	OSPF Authentication Type (0-2)
Authentication	8 Bytes	Determined by AuType
Network Mask	4 Bytes	Sending interface's netmask
Hello Interval	2 Bytes	OSPF Hello Interval
Options	1 Byte	
Rtr Pri	1 Byte	Interface OSPF Priority
Router Dead Interval	4 Bytes	OSPF Dead Interval
Designated Router	4 Bytes	OSPF DR IP Address
Backup Designated Router	4 Bytes	OSPF BDR IP Address
Neighbor	varies	Known Neighbor RID(s)

OSPF Hello packets are used for neighbor discovery and maintenance. OSPF routers use it to learn each other's RID in neighbor state **INIT** and determine whether they are OSPF-compatible.

OSPF routers are compatible if their RIDs are unique and if their **Version**, **Area ID**, **Network Mask**, **Hello Interval**, and **Router Dead Interval** match. If OSPF authentication is configured, the **AuType** and **Authentication** must also match. Two OSPF-compatible routers enter neighbor state **2WAY**, are considered OSPF neighbors, and are ready to begin the 2-way LSDB exchange process.

OSPF elections may take place among OSPF routers connected to the same subnet. The router with the highest **Rtr Pri** is elected *Designated Router* (DR), with the highest RID serving as tiebreaker; the runner-up becomes the *Backup Designated Router* (BDR). There is no preemption and the remaining routers become *DROthers*. The DR/BDR routers become fully-adjacent neighbors with all OSPF routers in the subnet; DROther routers remain in 2WAY state with each other.

Table 3.16: OSPF Database Description (DD)

Version	1 Byte	OSPF Version (2)
Type	1 Byte	OSPF Packet Type (2)
Packet Length	2 Bytes	Header + Packet Length in Bytes
Router ID	4 Bytes	Advertising Router RID
Area ID	4 Bytes	Advertising Router's Area ID
Checksum	2 Bytes	
AuType	2 Bytes	OSPF Authentication Type (0-2)
Authentication	8 Bytes	Determined by AuType
Interface MTU	2 Bytes	IP MTU in Bytes
Options	1 Byte	
Reserved	5 bits	0
Initial	1 bit	1 indicates first DD packet
More	1 bit	1 indicates more DD packets follow
Master/Slave	1 bit	1 indicates master, 0 indicates slave
DD Sequence Number	4 Bytes	sequence number of DD packets
LSA Headers	varies	LSA header(s) (Table 3.14)

OSPF DD packets are used by both routers to summarize and compare their local LSDBs. The *master router* (higher RID) enters neighbor state **EXSTART** by sending an OSPF DD packet, which initializes the LSDB exchange process. The *slave router* (lower RID) enters neighbor state **EXCHANGE** by responding with its own OSPF DD packet.

Table 3.17: OSPF Link-State Request (LSR)

Version	1 Byte	OSPF Version (2)
Type	1 Byte	OSPF Packet Type (3)
Packet Length	2 Bytes	Header + Packet Length in Bytes
Router ID	4 Bytes	Advertising Router RID
Area ID	4 Bytes	Advertising Router's Area ID
Checksum	2 Bytes	
AuType	2 Bytes	OSPF Authentication Type (0-2)
Authentication	8 Bytes	Determined by AuType
LS Type	4 Bytes	Requested LSA Type (Table 3.13)
Link-State ID	4 Bytes	Determined by LS Type
Advertising Router	4 Bytes	Sending Router's RID

Table 3.18: OSPF Link-State Update (LSU)

Version	1 Byte	OSPF Version (2)
Type	1 Byte	OSPF Packet Type (4)
Packet Length	2 Bytes	Header + Packet Length in Bytes
Router ID	4 Bytes	Advertising Router RID
Area ID	4 Bytes	Advertising Router's Area ID
Checksum	2 Bytes	
AuType	2 Bytes	OSPF Authentication Type (0-2)
Authentication	8 Bytes	Determined by AuType
Number of LSAs	4 Bytes	
LSAs	varies	LSA(s)

Table 3.19: OSPF Link-State Acknowledgment (LSAck)

Version	1 Byte	OSPF Version (2)
Type	1 Byte	OSPF Packet Type (5)
Packet Length	2 Bytes	Header + Packet Length in Bytes
Router ID	4 Bytes	Advertising Router RID
Area ID	4 Bytes	Advertising Router's Area ID
Checksum	2 Bytes	
AuType	2 Bytes	OSPF Authentication Type (0-2)
Authentication	8 Bytes	Determined by AuType
LSA Headers	varies	Acknowledged LSA header(s) (Table 3.14)

Both routers enter neighbor state **LOADING** after the LSDB exchange process has been initiated. The master router requests certain LSAs using OSPF LSR packets (Table 3.17). The slave router responds with OSPF LSU packets (Table 3.18) containing the requested LSAs, which the master router acknowledges using OSPF LSAck packets (Table 3.19). This process then reverses, with the slave router requesting LSAs that the master router supplies. Routers which complete the LSDB exchange process enter neighbor state **FULL** and are *fully-adjacent OSPF neighbors*.

OSPF runs *Dijkstra's Shortest-Path First* (SPF) algorithm on the LSDB to calculate the best routes to each network. This calculation depends on the LSDB size and is performed independently for each OSPF area. Each OSPF interface calculates an *OSPF Cost* ($OSPF\ Reference\ Bandwidth \div Interface\ Bandwidth$) which affects the resulting OSPF route metric.

4 Layer 4 Protocols

Table 4.1: Well-Known Ports

TCP	20	FTP Data
TCP	21	FTP Control
TCP	22	SSH / SFTP
TCP	23	Telnet
TCP	25	SMTP
TCP	49	Cisco TACACS+
	53	DNS
UDP	67	DHCP Server
UDP	68	DHCP Client
UDP	69	TFTP
TCP	80	HTTP
TCP	110	POP3
UDP	123	NTP
UDP	161	SNMP Agent
UDP	162	SNMP Manager
TCP	443	HTTPS
UDP	514	Syslog
TCP	989	FTPS Data
TCP	990	FTPS Control

Table 4.2: Registered Ports

UDP	1812	RADIUS Auth
UDP	1813	RADIUS Accounting
TCP	1985	Cisco HSRP
UDP	5246	CAPWAP Control
UDP	5247	CAPWAP Data

Note: *Well-Known Ports* range from 0-1023, *Registered Ports* range from 1024-49151, and *Ephemeral Ports* range from 49152-65535.

4.1 RFC 768 User Datagram Protocol (UDP)

Source Port	2 Bytes	Tables 4.1 and 4.2
Destination Port	2 Bytes	Tables 4.1 and 4.2
Length	2 Bytes	Header + Data Length in Bytes
Checksum	2 Bytes	
Data	varies	

UDP is a *connectionless* protocol offering port-based multiplexing without any delivery guarantee. This is optimal for latency-sensitive applications which are fault-tolerant or implement their own error recovery mechanisms, such as VoIP, DNS, DHCP, TFTP, SNMP, Syslog, QUIC, and others.

4.2 RFC 793 Transmission Control Protocol (TCP)

Source Port	2 Bytes	Tables 4.1 and 4.2
Destination Port	2 Bytes	Tables 4.1 and 4.2
SEQ Number	4 Bytes	Windowing / Flow Control
ACK Number	4 Bytes	Windowing / Flow Control
Data Offset	4 bits	
Reserved	4 bits	Future Use (0)
Flags	8 bits	Connection Management
Window	2 Bytes	Connection Window Size
Checksum	2 Bytes	
Urgent	2 Bytes	Last urgent data byte
Options	varies	≤ 40 Bytes of Optional Fields
Data	varies	

TCP is a *connection-oriented* protocol offering **error recovery**, **flow control**, **connection establishment/termination**, and **ordered data transfer / segmentation** features at the cost of additional overhead.

The TCP *3-way handshake* (SYN SYN-ACK ACK) sets up the connection SEQ Number and ACK Number for ordered data transfer. The Window determines how much data can be sent before requiring an acknowledgement. It increases when no errors occur and decreases for each missed acknowledgement, providing flow control and error recovery. Data is assumed lost/errored if not acknowledged within a certain time interval. The TCP *4-way termination* process (FIN-ACK ACK FIN-ACK ACK) closes the connection after data transfer is complete.

5 Layer 5 Applications

5.1 RFC 959 File Transfer Protocol (FTP)

FTP manages file transfers between FTP clients and FTP servers in clear-text over TCP. A persistent *FTP Control Connection* over port 21 defines the available FTP client functions for remotely managing/transferring files. An *FTP Data Connection* is established only when a file is transferred and often uses a nonstandard port. It is established by the server in *FTP Active Mode*, or by the client in *FTP Passive Mode*; passive mode is required when the client sits behind a firewall or NAT.

Note: The client issues the FTP PORT command for FTP Active mode and the FTP PASV command for FTP Passive mode. The client issues the FTP AUTH command for FTPS Explicit mode.

FTP Secure (FTPS) implements TLS to provide secure authentication and data encryption. FTPS clients using *FTPS Explicit Mode* must explicitly initialize a TLS tunnel after establishing each

FTP connection. FTPS clients using *FTPS Implicit Mode* begin each FTP connection with a TLS connection automatically.

SSH FTP (SFTP) encrypts file transfers over an SSH connection instead of using TLS. *Trivial FTP* (TFTP) uses UDP with built-in error detection and offers minimal functionality.

5.2 RFC 2131 Dynamic Host Configuration Protocol (DHCP)

Message Type	1 Byte	Message Type (request/reply)
HTYPE	1 Byte	L2 Protocol (1 for Ethernet)
HLEN	1 Byte	L2 Address Length in Bytes
Hops	1 Byte	Optionally boot via relay agent
Transaction ID	4 Bytes	Identifies client-server exchange (random)
Seconds	2 Bytes	Time since client began request
Broadcast Flag	1 bit	broadcast/unicast replies
Reserved	15 bits	Future Use (0)
Client IP	4 Bytes	Current Client IP
Your IP	4 Bytes	Your Client IP
Next Server IP	4 Bytes	Next server in bootstrap process
Relay Agent IP	4 Bytes	Boot via relay agent
CHADDR	16 Bytes	Client Hardware Address
Server Name	64 Bytes	Optional Server Hostname
File Name	128 Bytes	Optional Boot File
Magic Cookie	4 Bytes	Identifies DHCP over Bootp (0x63.82.53.63)
Options	varies	One or more option headers (Table 5.1)

Table 5.1: DHCP Options

Option	Value
1	Subnet Mask
3	Default Router
6	DNS Server
12	Client Hostname
28	Broadcast Address
43	WLC IP Address
50	Requested IP Address
51	Address Lease Time
53	DHCP Message Type
54	DHCP Server ID
55	Parameter Requests
58	Renewal Time Value
59	Rebinding Time Value
66	TFTP Server IP Address
82	Relay Agent Information
150	TFTP Server List
255	End

The DHCP Lease Process:

1. The DHCP client broadcasts a **DHCP Discover** message to any local DHCP servers.
2. The DHCP server replies with a **DHCP Offer** message containing configuration information.
3. The client broadcasts a **DHCP Request** message to accept or a **DHCP Decline** message deny an offer.
4. The server replies to a successful request with a **DHCP Acknowledge** message.
5. The client can send a **DHCP Release** message any time to release its current DHCP lease.

DHCP Relay can be configured on the default gateway LAN interface when the DHCP server exists in another subnet. This causes the default gateway to insert **DHCP Option 82** into received DHCP client messages and forward them to the DHCP server as unicast packets. *DHCP Snooping*

automatically enables this feature as the *DHCP Snooping Information Option*.

DHCP Snooping is an optional switch security feature to prevent DHCP Poisoning and DHCP DoS. The *DHCP Snooping Binding Table* maps DHCP **CHADDR** values to their corresponding switchport, VLAN, frame **Source MAC**, and packet **Source IP** values based on observed DHCP flows. It ensures consistency between table entries and subsequent DHCP client messages; all DHCP server messages are filtered by default. This behavior is disabled on DHCP Snooping *trusted ports*. DHCP Snooping uses optional per-interface rate limits to prevent DoS attacks against the switch CPU and DHCP servers.

5.3 RFC 7230 HyperText Transfer Protocol (HTTP)

Note: See RFCs 9110/9111 plus the IANA Field Name Registry and IANA Message Headers for HTTP header fields and their names.

HTTP manages file transfers between web clients (browsers) and web servers over TCP. Client requests specify a *Uniform Resource Identifier* (URI, RFC 7595) in **SCHEME://AUTHORITY/PATH** format (typically **PROTOCOL://SERVER_NAME/OBJECT** for web pages). Requests also specify an *HTTP Verb*, typically corresponding to a *CRUD* action. Web servers respond to client requests with an *HTTP Return Code* and any relevant data.

Table 5.2: Common HTTP Verbs

HTTP Verb	CRUD Action	Description
HTTP POST	Create	Create/initialize new data structures/variables
HTTP GET	Read	Read variable names/structures/values
HTTP PATCH/PUT	Update	Update/replace some variable values
HTTP DELETE	Delete	Delete some data structures/variables

Table 5.3: HTTP Return Codes

Code	Type	Meaning
1XX	Informational	Request was received and is being processed
2XX	Successful	Request was received, understood, and accepted
3XX	Redirection	Further action required to complete request
4XX	Client Error	Request contains bad syntax or cannot be fulfilled
5XX	Server Error	Server failed to fulfill an (apparently) valid request

Modern web browsers support *HTTP Secure* (HTTPS) using TLS (RFC 5246) to dynamically secure a connection between the client and server. Initializing a TLS session creates a client VPN tunnel which authenticates the user and encrypts the data transfer between that client and server only.

5.4 RFC 5424 System Logging (Syslog)

Cisco IOS generates system event logs categorized based on *severity level* (Table 5.4). Log messages use the **TIMESTAMP SEQ_NUM %FACILITY-SEVERITY-MNEMONIC: DESCRIPTION** format and may be stored locally for review. Devices can be configured for centralized logging to a Syslog server over UDP based on severity level, freeing local device RAM for other tasks and simplifying event correlation among devices.

Table 5.4: Logging Keywords

Keyword	Severity	Description
Emergency	0	System Unusable
Alert	1	Immediate Action Required
Critical	2	Critical Conditions
Error	3	Error Conditions
Warning	4	Warning Conditions
Notification	5	Significant Conditions
Informational	6	Information
Debug	7	debug Messages

6 RFC 2475 Quality of Service (QoS)

PCP7	CS7			
<i>Network Control</i>	56 0x38			
PCP6	CS6			
<i>Internetwork Control</i>	48 0x30			
PCP5	CS5			EF
<i>Voice</i>	40 0x28			46 0x2e
PCP4	CS4	AF41	AF42	AF43
<i>Video</i>	32 0x20	34 0x22	36 0x24	38 0x26
PCP3	CS3	AF31	AF32	AF33
<i>Critical Apps</i>	24 0x18	26 0x1a	28 0x1c	30 0x1e
PCP2	CS2	AF21	AF22	AF23
<i>Excellent Effort</i>	16 0x10	18 0x12	20 0x14	22 0x16
PCP1	CS1	AF11	AF12	AF13
<i>Background</i>	8 0x08	10 0x0a	12 0x0c	14 0x0e
PCP0	CS0			
<i>Best Effort</i>	0 0x00			

Note: CS values use bit-pattern XXX000, resulting in an $8X$ DSCP value. AF values use bit-pattern XXXYY0, resulting in an $8X + 2Y$ DSCP value.

Table 6.1: QoS Fields

QoS Standard	Protocol Field(s)	Notes
IEEE 802.1p (CoS)	802.1Q PCP/DEI	Table 2.5
	802.11 QoS Control	Section 2.7
RFC 791 IPP (CS)	IPv4 ToS Byte (OLD)	Section 3.1
RFC 2474/3168 (DSCP)	IPv4 DS Field (NEW)	
RFC 2460 (DSCP)	IPv6 Traffic Class	Section 3.2
RFC 3032/5462	MPLS EXP	Section 7

Quality of Service (QoS) is defined across several standards to provide preferential treatment for certain traffic. They define groups of QoS markings (shown above) as well as header fields used to mark QoS traffic. QoS-configured devices employ *Classification* to identify QoS traffic and apply specifically configured actions, including *Marking*, *Queuing*, *Policing*, *Shaping*, and/or *Congestion Avoidance* features. Marking is typically done by an initial device, simplifying Classification for downstream devices. QoS characterizes network traffic according to the following metrics:

Bandwidth: The speed of a link in *bits-per-second* (bps), or the capacity of the link to send a number of bits per-second.

One-Way Delay: The time between sending one packet and that same packet arriving at the destination host.

Round-Trip Delay: One-way delay plus the time for the receiver to send a packet back; the time to send one packet between two hosts and receive one back.

Jitter: The variation in one-way delay between any consecutive packets sent by the same application.

Loss: The number of lost messages expressed as a percentage of sent packets.

Certain types of traffic have strict requirements for these metrics, which must be met in order to ensure a good end-user experience. Technical limitations may also require much higher bandwidth per-flow for certain traffic (e.g. video).

Table 6.2: Traffic QoS Requirements

Traffic	One-Way Delay	Jitter	Loss
<i>Voice Over IP</i> (VoIP)	< 150 ms	< 30 ms	< 1%
<i>Video</i>	200-400 ms	30-50 ms	0.1-1%

Note: A majority of loss occurs due to normal network operations, but may involve faulty hardware or network congestion.

7 RFC 3031 Multiprotocol Label Switching (MPLS)

Table 7.1: MPLS WAN Services

MPLS Service	Nodes	Notes
Layer 2 E-Line	2 per-EVC	P2P EVCs using 802.1Q VLAN IDs and subnets
Layer 2 E-LAN	n per-EVC	Mesh EVC using 1 subnet
Layer 2 E-Tree	n per-EVC	P2MP EVC using 1 subnet
Layer 2 MPLS VPN	n	CE-CE neighbors within 1 subnet
Layer 3 MPLS VPN	n	PE-CE neighbors with route redistribution

7.1 MPLS VPNs

A *Service Provider* (SP) connects customers to their *MPLS network* via *access links*. Each access link connects *Provider Edge* (PE) and *Customer Edge* (CE) routers. PE routers replace incoming frames' Data-Link headers, supporting many Internet access technologies (serial, MetroE, 4G/5G wireless, CATV/DSL, etc). They also insert/remove an MPLS header between Layers 2 and 3 for forwarding within the MPLS network. MPLS *label switching* logic isolates customer IP routes and packets from other customers, creating an unencrypted private *MPLS VPN* without fear of eavesdropping.

Layer 2 MPLS VPNs are transparent to the customer; all CE routers belong to the same subnet and become routing peers. Layer 2 MPLS VPNs include *Ethernet over MPLS* (EoMPLS, RFC 4448), *Virtual Private Wire Service* (VPWS, RFC 6624), and *Virtual Private LAN service* (VPLS, RFCs 4761/4762).

Layer 3 MPLS VPNs (RFCs 4364/4577) provide Layer 3 IP routing and QoS services. Customers must share IP addressing plans and mark QoS traffic sent across the access link. PE and CE routers

become routing peers, allowing the SP to perform route redistribution via MP-BGP. This ensures the correct customer routes are advertised to customer sites and allows the MPLS network to make the necessary traffic forwarding decisions.

7.2 RFCs 6004/7387 Metro Ethernet (MetroE) WANs

MetroE encompasses a set of common Ethernet WAN services defined by the MEF. SPs use MPLS networks and EoMPLS/VPWS/VPLS Layer 2 VPNs to build and maintain MetroE services. Customers connect to a local SP *Point-of-Presence* office via an Ethernet access link. A *User Network Interface* (UNI, RFC 6005) defines the standards used by the access link.

MetroE services operate at Layer 2 and are transparent to the customer; all CE equipment belongs to the same subnet and become routing peers. SP equipment forwards customer Ethernet frames based on their **Source/Destination MAC** and **802.1Q VLAN ID** values. Customers must choose an *Ethernet Virtual Connection* (EVC), which determines the scope of customer site communication.

An *Ethernet Line* (E-Line) connects exactly two customer sites in a Point-to-Point EVC. These customer devices must belong to the same subnet to become routing peers. Multiple E-Lines can terminate at the same customer interface, each using a unique subnet and 802.1Q VLAN ID. This enables many E-Lines to share the same access link.

An *Ethernet LAN* (E-LAN) connects multiple customer sites in a Full Mesh EVC. These customer devices must reside in the same subnet to become routing peers. An E-LAN is much more practical than using $\frac{n(n-1)}{2}$ separate E-Lines.

An *Ethernet Tree* (E-Tree) connects an *E-Tree Root* (a central site) to multiple *E-Tree Leafs* (remote sites) in a Point-to-Multipoint EVC. Each E-Tree leaf can send frames only to the E-Tree root, while the E-Tree root can send frames to all E-Tree leafs. This is optimal when an E-LAN is not required.

8 Appendix

Table 8.1: Protocol Timers

Protocol	Timer	Default Value
MAC Table	Aging-Time	300 s
	Hello	2 s
STP	Forward Delay	15 s
	Max Age	20 s (10* Hello)
RSTP	Hello	2 s
	Max Age	6 s (3* Hello)
CDP	Update	60 s
	Hold	180 s (3* Update)
LLDP	Send	30 s
	Hold	120 s (4* Send)
Errdisable	Recovery	300 s
Port Security	Aging	300 s
	Hello	10 s
	Dead	40 s (4* Hello)
OSPF	LSA Age	30 mins per-LSA

Table 8.2: Protocol Priority Values

Protocol	Priority	Default Value
STP/RSTP	0 - 61,440	32,768
OSPF	0 - 255	1
HSRP	1 - 255	100
VRRP	1 - 254	100
NTP	1 - 15	8

Note: The MAC Address Table is also called the *Bridging* table or *Content Addressable Memory* (CAM) table.

Table 8.3: Cisco Encryption Algorithms

Type	Algorithm	Salt	Secure?
0	cleartext		NO
4	PBKDF2-SHA-256		NO
5	MD5	32 bits	WEAK
6	AES-128		YES
7	Vigenere		NO
8	PBKDF2-SHA-256	80 bits	YES
9	Scrypt	80 bits	YES

Table 8.4: ASCII Values

32	64	96	0x20	0x40	0x60	SP	@	‘
33	65	97	0x21	0x41	0x61	!	A	a
34	66	98	0x22	0x42	0x62	“	B	b
35	67	99	0x23	0x43	0x63	#	C	c
36	68	100	0x24	0x44	0x64	\$	D	d
37	69	101	0x25	0x45	0x65	%	E	e
38	70	102	0x26	0x46	0x66	&	F	f
39	71	103	0x27	0x47	0x67	‘	G	g
40	72	104	0x28	0x48	0x68	(	H	h
41	73	105	0x29	0x49	0x69	)	I	i
42	74	106	0x2a	0x4a	0x6a	*	J	j
43	75	107	0x2b	0x4b	0x6b	+	K	k
44	76	108	0x2c	0x4c	0x6c	,	L	l
45	77	109	0x2d	0x4d	0x6d	-	M	m
46	78	110	0x2e	0x4e	0x6e	.	N	n
47	79	111	0x2f	0x4f	0x6f	/	O	o
48	80	112	0x30	0x50	0x70	0	P	p
49	81	113	0x31	0x51	0x71	1	Q	q
50	82	114	0x32	0x52	0x72	2	R	r
51	83	115	0x33	0x53	0x73	3	S	s
52	84	116	0x34	0x54	0x74	4	T	t
53	85	117	0x35	0x55	0x75	5	U	u
54	86	118	0x36	0x56	0x76	6	V	v
55	87	119	0x37	0x57	0x77	7	W	w
56	88	120	0x38	0x58	0x78	8	X	x
57	89	121	0x39	0x59	0x79	9	Y	y
58	90	122	0x3a	0x5a	0x7a	:	Z	z
59	91	123	0x3b	0x5b	0x7b	;	[	{
60	92	124	0x3c	0x5c	0x7c	<	\	
61	93	125	0x3d	0x5d	0x7d	=	]	}
62	94	126	0x3e	0x5e	0x7e	>	^	~
63	95	127	0x3f	0x5f	0x7f	?	_	DEL

Table 8.5: SI Prefixes

Z	Zetta	10 ²¹	sextillion
E	Exa	10 ¹⁸	quintillion
P	Peta	10 ¹⁵	quadrillion
T	Tera	10 ¹²	trillion
G	Giga	10 ⁹	billion
M	Mega	10 ⁶	million
k	Kilo	10 ³	thousand
h	Hecto	10 ²	hundred
da	Deka	10 ¹	ten
d	deci	10 ⁻¹	tenth
c	centi	10 ⁻²	hundredth
m	milli	10 ⁻³	thousandth
μ	micro	10 ⁻⁶	millionth
n	nano	10 ⁻⁹	billionth
p	pico	10 ⁻¹²	trillionth

Table 8.6: Binary Prefixes

Yi	Yobi	2 ⁸⁰
Zi	Zebi	2 ⁷⁰
Ei	Exbi	2 ⁶⁰
Pi	Pebi	2 ⁵⁰
Ti	Tebi	2 ⁴⁰
Gi	Gibi	2 ³⁰
Mi	Mebi	2 ²⁰
Ki	Kibi	2 ¹⁰

Table 8.7: IEEE 802.1x/IEEE 802.11i Extensible Authentication Protocol (EAP)

Authentication	Open Authentication (Open)		802.11 compliance
	Wired Equivalent Privacy (WEP)		Static WEP Keys
	IEEE 802.1x / 802.11i EAP	Lightweight EAP (LEAP)	Dynamic WEP Keys
		EAP Flexible Authentication by Secure Tunneling (EAP-FAST)	Protected Access Credential (PAC)
		Protected EAP (PEAP)	Digital Certificate (AS)
		EAP Transport Layer Security (EAP-TLS)	Digital Certificate (AS + Client)
Privacy and Integrity	Temporal Key Integrity Protocol (TKIP)		WPA Personal WPA Enterprise
	AES Counter/CBC-MAC Protocol (CCMP)		WPA2 Personal WPA2 Enterprise
	AES Galois/Counter Mode Protocol (CCMP)		WPA3 Personal WPA3 Enterprise

8.1 Official Standards

OFFICIAL STANDARD	ALTERNATIVE	REFERENCE
IEEE 802.3 Copper	ANSI/TIA-598 Fiber	Section 1.2
IEEE 802.3af/at/bt PoE	Cisco ILP	Table 1.4
Ethernet II / IEEE 802.3		Section 2.1
IEEE 802.2 LLC/SNAP		Tables 2.3 and 2.4
IEEE 802.1Q VLAN Tagging	Cisco ISL / DTP / VTP	Sections 2.2 and 2.3
IEEE 802.1AB LLDP	Cisco CDP	Section 2.4
IEEE 802.1D/w STP/RSTP	Cisco PVST+/PVRST	Section 2.5
IEEE 802.1s MSTP		
IEEE 802.3AD LACP	Cisco PAgP	
RFC 826 ARP	RFC 4861 NDP	Sections 2.6 and 3.4
IEEE 802.11 WLANs	Wi-Fi Alliance	Section 2.7
IEEE 802.11i EAP		Table 8.7
IEEE 802.1x Access Control		Table 8.7
ITU HDLC / cHDLC	RFC 1661 PPP	Sections 2.8 and 2.9
RFC 791/1918 IPv4	RFC 2460 IPv6	Sections 3.1 and 3.2
RFC 792 ICMP	RFC 4443 ICMPv6	Section 3.3
RFC 2328 OSPFv2	EIGRP / RIP / BGP / ...	Section 3.5
RFC 5798 VRRP	Cisco HSRP / GLBP	
RFC 1631/3022 NAT		
RFC 2475 QoS		Section 6
RFC 3031 MPLS		Section 7
RFC 768 UDP	RFC 793 TCP	Sections 4.1 and 4.2
RFC 959 FTP	FTPS / TFTP / SFTP / ...	Section 5.1
RFC 2131 DHCP	RFC 3046 DHCP Relay	Section 5.2
RFC 7230 HTTP	HTTPS	Section 5.3
RFC 5424 Syslog		Section 5.4
RFC 1065 SNMP		
RFC 1305/5905 NTP		
RFC 4301 IPSec	RFC 5246 TLS	

8.2 Cisco IOS Configuration Examples

Layer 2 Interface Configuration

```
configure terminal
  interface F0/1
    mac-address MAC
    bandwidth KBPS
    duplex {half | full | auto}
    speed {MBPS | auto}
    description TEXT

show interfaces [INT] [status | switchport]
```

VLAN Configuration

```
configure terminal
  vlan VLAN_ID
    name TEXT
    [no] shutdown
  [no] shutdown vlan VLAN_ID
  interface F0/1
    switchport trunk encapsulation {dot1Q | isl | negotiate}
    switchport mode {trunk | dynamic {desirable | auto}}
    switchport nonegotiate
    switchport trunk allowed vlan VLAN_LIST
    switchport trunk native vlan VLAN_ID
    [no] shutdown
  interface range F0/2 - 12
    switchport mode access
    switchport {access | voice} vlan VLAN_ID
    [no] shutdown
  vtp mode {server | client | transparent | off}
  vtp domain TEXT
  vtp password PASSWORD
  [no] vtp pruning
  vtp version 2

show interfaces [INT] [status | switchport | trunk]
show vlan brief
show vlan id VLAN_ID
show vtp {status | password}
```

CDP / LLDP Configuration

```
configure terminal
  [no] {cdp | lldp} run
  {cdp | lldp} timer SECS
  {cdp | lldp} holdtime SECS
  [no] cdp advertise-v2
  interface range F0/2 - 12
    [no] cdp enable
    [no] lldp {transmit | receive}

show {cdp | lldp}
show {cdp | lldp} traffic
show {cdp | lldp} interface [INT]
show {cdp | lldp} neighbors [detail] [INT]
show {cdp | lldp} entry NEIGHBOR
```

Spanning Tree Configuration

```
configure terminal
  errdisable recovery cause bpduguard
  spanning-tree mode {pvst | rapid-pvst | mst}
  spanning-tree pathcost method {long | short}
  spanning-tree [vlan VLAN_ID] root {primary | secondary}
  spanning-tree [vlan VLAN_ID] priority {32768 | 28672 | 24576 | ...}
  spanning-tree portfast [edge | network] [bpduguard | bpdufilter] default
  spanning-tree loopguard default
interface Po1
  spanning-tree [vlan VLAN_ID] cost PORT_COST
  spanning-tree [vlan VLAN_ID] port-priority PORT_PRIO
  spanning-tree [vlan VLAN_ID] link-type {point-to-point | shared}
  spanning-tree portfast [disable | [edge | network] [default | trunk]]
  spanning-tree {bpduguard | bpdufilter} {enable | disable}
  spanning-tree guard {root | loop | none}

show spanning-tree [bridge | summary]
show spanning-tree [vlan VLAN_LIST | interface INT]
```

Link Aggregation Configuration

```
configure terminal
  port-channel load-balance {src-mac | dst-mac | src-dst-mac | src-ip | dst-ip
    > | src-dst-ip | src-port | dst-port | src-dst-port}
interface range F0/13 - 16
  [no] switchport
  [no] shutdown
  channel-group 1 mode {on | desirable | auto | active | passive}
interface Po1
  [no] switchport
  [no] shutdown

show etherchannel [CHANNEL] {summary | port-channel}
show etherchannel load-balance
test etherchannel load-balance interface INT mac SRC_MAC DST_MAC
```

Authenticated NTP Configuration

```
configure terminal
  clock timezone CST -6 0
  clock summer-time CDT recurring 2 SUNDAY MAR 02:00 1 SUNDAY NOV 02:00
clock set HH:MM:SS DATE MONTH YEAR
clock {update-calendar | read-calendar}
configure terminal
  ntp authenticate
  ntp authentication-key 1 md5 PASSWORD
  ntp trusted-key 1
  ntp master STRATUM
  ntp {peer | server} {A.B.C.D | HOSTNAME} [key 1] [prefer]
  ntp update-calendar
  ntp source loopback 0

show ntp status
show ntp associations [detail]
show {clock | calendar} [detail]
```

Logging and SNMP Configuration

```

terminal monitor
configure terminal
  logging console {0-7 | emergency | alert | critical | error | warning |
    > notification | informational | debug}
  logging monitor {0-7 | emergency | alert | critical | error | warning |
    > notification | informational | debug}
  logging buffered [MEMSIZE] {0-7 | emergency | alert | critical | error |
    > warning | notification | informational | debug}
  logging [host] {A.B.C.D | HOSTNAME}
  logging trap {0-7 | emergency | alert | critical | error | warning |
    > notification | informational | debug}
[no] service {timestamps | sequence-numbers}
logging source-interface INT
snmp-server community PASSWORD {ro | rw}
snmp-server {contact | location | chassis-id} TEXT
snmp-server host {A.B.C.D | HOSTNAME} [trap | inform] version 2c PASSWORD
snmp-server enable traps TRAPS_LIST

{show | clear} logging
show snmp {community | contact | location | host}

```

Port Security Configuration

```

configure terminal
  errdisable recovery cause psecure-violation
  errdisable recovery interval SECS
interface range F0/2 - 12
  switchport mode {access | trunk}
  switchport port-security violation {protect | restrict | shutdown}
  switchport port-security maximum MAX
  switchport port-security mac-address {MAC | sticky}
  switchport port-security aging type {absolute | inactivity}
  switchport port-security aging time MINS
  switchport port-security aging static
  switchport port-security
  mac address-table aging-time SECS [vlan VLAN_ID]

show errdisable recovery
show interfaces [INT] [status]
show port-security [interface INT]
show mac address-table [aging-time | count | learning]
show mac address-table [dynamic | secure | static] [address MAC | interface INT
  > | vlan VLAN_ID]
clear mac address-table dynamic [vlan VLAN_ID | interface INT | address MAC]

```

DHCP Snooping and DAI Configuration

```

configure terminal
  errdisable recovery cause dhcp-rate-limit
  errdisable recovery cause arp-inspection
ip dhcp snooping
ip dhcp snooping vlan VLAN_LIST
[no] ip dhcp snooping information option
ip arp inspection vlan VLAN_LIST
ip arp inspection validate {[src-mac] [dst-mac] [ip]}
interface Po1
  ip dhcp snooping trust
  ip arp inspection trust

```

```

interface F0/1
  ip arp inspection trust
interface range F0/2 - 12
  ip dhcp snooping limit rate MAX
  ip arp inspection limit rate MAX [burst-interval SECS]

show ip dhcp snooping [binding]
show ip arp inspection [statistics | interfaces]

```

Layer 3 Interface Configuration

```

configure terminal
  sdm prefer lanbase-routing
  ip routing
  ipv6 unicast-routing
  interface F0/0.SUBINT
    encapsulation dot1q VLAN_ID [native]
    ip address {A.B.C.D M.M.M.M | dhcp}
    {ipv6 enable | ipv6 address {ADDRESS/PREFIX_LENGTH [link-local | anycast] |
      > PREFIX/64 eui-64 | dhcp | autoconfig}}
    [no] shutdown
  interface vlan VLAN_ID
    ip address {A.B.C.D M.M.M.M | dhcp}
    {ipv6 enable | ipv6 address {ADDRESS/PREFIX_LENGTH [link-local | anycast] |
      > PREFIX/64 eui-64 | dhcp | autoconfig}}
    [no] shutdown
  interface F0/1
    [no] switchport
    ip address {A.B.C.D M.M.M.M | dhcp}
    {ipv6 enable | ipv6 address {ADDRESS/PREFIX_LENGTH [link-local | anycast] |
      > PREFIX/64 eui-64 | dhcp | autoconfig}}
    [no] shutdown
    ip helper-address DHCP_SERVER
  interface S0/0/0
    encapsulation {hdlc | frame-relay | ppp}
    ppp authentication {[chap] [pap]}
    clock rate BPS
    bandwidth KBPS
    ip address {A.B.C.D M.M.M.M | dhcp}
    {ipv6 enable | ipv6 address {ADDRESS/PREFIX_LENGTH [link-local | anycast] |
      > PREFIX/64 eui-64 | dhcp | autoconfig}}
    [no] shutdown

show sdm prefer
show {ip | ipv6} interface [brief | INT]
show interfaces [INT] [switchport | trunk]
show controllers [INT]
show protocols [INT]
show dhcp lease
show ip default-gateway
show vlans

```

IP Routing Configuration

```

configure terminal
  router ospf 1
    router-id {A.B.C.D | VALUE}
    auto-cost reference-bandwidth MBPS
    maximum-paths 4

```



```

distance 110
default-information originate [always]
[no] passive-interface {INT | default}
[no] network A.B.C.D W.W.W.W area AREA
[no] shutdown
area AREA authentication message-digest
key chain NAME
key KEY_ID
key-string PASSWORD
cryptographic-algorithm {hmac-sha-1 | hmac-sha-256 | hmac-sha-384 |
    > hmac-sha-512 | md5}
send-lifetime START_TIME {infinite | END_TIME | duration SECS}
interface S0/0/0
ip ospf 1 area AREA
ip ospf network {point-to-point | broadcast}
ip ospf cost PORT_COST
ip ospf priority 0-255
ip ospf hello-interval SECS
ip ospf dead-interval SECS
ip ospf message-digest-key 1 md5 PASSWORD
ip ospf authentication {message-digest | key-chain NAME}
router rip
version 2
no auto-summary
[no] network NETWORK_ID
[no] passive-interface INT
default-information originate
maximum-paths VALUE
distance AD
[no] shutdown
router eigrp AS_VALUE
eigrp router-id A.B.C.D
no auto-summary
[no] network A.B.C.D [W.W.W.W]
[no] passive-interface INT
default-information originate
maximum-paths VALUE
variance VALUE
distance INTERNAL_AD EXTERNAL_AD
[no] shutdown
ip route A.B.C.D M.M.M.M {[EXIT_INT] [NEXT_HOP]} [AD] [permanent]
ipv6 route PREFIX/LENGTH {[EXIT_INT] [NEXT_HOP]} [AD] [permanent]

show {ip | ipv6} protocols
show ip ospf
show ip ospf interface [INT | brief]
show ip ospf neighbor
show ip ospf database
clear ip ospf [PROCESS_ID] process
show key chain NAME
show {ip | ipv6} route [connected | local | static | ospf | ...] [ADDR]
show ip arp
show ipv6 neighbors

```

VRF Configuration

```

configure terminal
ip vrf VRF_NAME
interface F0/0

```

```

ip vrf forwarding VRF_NAME
ip address {A.B.C.D M.M.M.M | dhcp}
[no] shutdown
show ip vrf
show ip route vrf VRF_NAME
ping vrf VRF_NAME [ADDRESS]

```

FHRP Configuration

```

configure terminal
interface F0/0
standby version 1-2
standby GROUP_ID ip A.B.C.D
standby GROUP_ID priority 1-255
standby GROUP_ID preempt
standby GROUP_ID description TEXT
vrrp GROUP_ID ip A.B.C.D [secondary]
vrrp GROUP_ID priority 1-254
vrrp GROUP_ID preempt [delay minimum SECS]
vrrp GROUP_ID description TEXT

show standby [brief]
show standby neighbors [INT]
show vrrp [brief | GROUP_ID]
show vrrp interface INT [brief]

```

ACL Configuration

```

configure terminal
access-list {1-99 | 1300-1999} {permit | deny} {[host] SRC_IP | SRC_IP SRC_WC
> | any} [log]
access-list {100-199 | 2000-2699} {permit | deny} {ip | icmp} {host SRC_IP |
> SRC_IP SRC_WC | any} {host DST_IP | DST_IP DST_WC | any} [log]
access-list {100-199 | 2000-2699} {permit | deny} {tcp | udp} {host SRC_IP |
> SRC_IP SRC_WC | any} [{eq | neq | lt | gt | range} SRC_PORT] {host
> DST_IP | DST_IP DST_WC | any} [{eq | neq | lt | gt | range} DST_PORT]
> [log]
access-list {1-199 | 1300-2699} remark TEXT
ip access-list standard {ACL_NAME | ACL_ID}
[no] [SEQ] {permit | deny} {[host] SRC_IP | SRC_IP SRC_WC | any} [log]
[no] [SEQ] remark TEXT
no SEQ
ip access-list extended {ACL_NAME | ACL_ID}
[no] [SEQ] {permit | deny} {ip | icmp} {host SRC_IP | SRC_IP SRC_WC | any}
> {host DST_IP | DST_IP DST_WC | any} [log]
[no] [SEQ] {permit | deny} {tcp | udp} {host SRC_IP | SRC_IP SRC_WC | any}
> [{eq | neq | lt | gt | range} SRC_PORT] {host DST_IP | DST_IP DST_WC |
> any} [{eq | neq | lt | gt | range} DST_PORT] [log]
[no] [SEQ] remark TEXT
no SEQ
interface S0/0/0
ip access-group {ACL_ID | ACL_NAME} {in | out}
line vty 0 15
access-class {ACL_ID | ACL_NAME} {in | out}
ip access-list resequence {ACL_ID | ACL_NAME} START STEP

show access-lists
show ip access-lists

```

Login Security Configuration

```
configure terminal
  hostname NAME
  ip domain-name NAME
  ip default-gateway A.B.C.D
  crypto key zeroize rsa
  crypto key generate rsa general-keys modulus LENGTH
  [no] ip ssh version {1 | 2}
  ip ssh time-out SECS
  ip ssh authentication-retries COUNT
  service password-encryption
  security passwords min-length LEN
  enable [algorithm-type {md5 | scrypt | sha256}] secret PASSWORD
  username NAME [algorithm-type {md5 | scrypt | sha256}] secret PASSWORD
  banner motd DELIMITER TEXT DELIMITER
  login block-for SECS attempts COUNT within SECS
  login quiet-mode access-class {ACL_ID | ACL_NAME}
  login delay SECS
  login {on-success | on-failure} log [every COUNT]
  security authentication failure rate THRESHOLD log
  line {vty 0 15 | console 0}
    [no] exec-timeout MINS [SECS]

show crypto key mypubkey rsa
show ip ssh
show login [failures]
```

AAA Configuration

```
configure terminal
  aaa new-model
  radius-server host A.B.C.D
  radius-server key PASSWORD
  tacacs-server host A.B.C.D [key PASSWORD] [timeout SECS] [port PORT]
    > [single-connection]
  tacacs-server key PASSWORD
  radius server NAME
    address ipv4 A.B.C.D [auth-port PORT] [acct-port PORT]
    key PASSWORD
    timeout SECS
  tacacs server NAME
    address ipv4 A.B.C.D
    key PASSWORD
    timeout SECS
    port PORT
    single-connection
  ip {radius | tacacs} source-interface INT
  aaa group server {radius | tacacs+} AAA_GROUP
    server {A.B.C.D [auth-port PORT] [acct-port PORT] | name NAME}
  aaa authentication login {default | AUTH_LIST} {[none] [local] [local-case]
    > [enable] [group {radius | tacacs+ | AAA_GROUP}]}
  aaa local authentication attempts max-fail COUNT
  line vty 0 15
    login authentication {default | AUTH_LIST}

{show | clear} aaa local user logout
show aaa user
show aaa sessions
```

QoS Configuration

```
configure terminal
  class-map [match-all | match-any] CMAP_NAME
    match protocol PROTOCOL
  policy-map PMAP_NAME
    class CMAP_NAME
      set ip dscp {EF | AFXY | CSX | BINARY}
      priority percent BANDWIDTH
      bandwidth percent BANDWIDTH
  interface F0/0
    service-policy {input | output} PMAP_NAME

show run | section policy-map
```

DHCP Services Configuration

```
configure terminal
  service dhcp
  ip dhcp excluded-address FIRST_IP [LAST_IP]
  ip dhcp pool POOL_NAME
    network A.B.C.D {M.M.M.M | /CIDR}
    domain-name TEXT
    default-router A.B.C.D
    dns-server A.B.C.D
    lease {DAYS HRS MINS | infinite}
    option 43 ip WLC_IP
    option 66 ip TFTP_IP
  ipv6 dhcp pool POOL_NAME
    address prefix PREFIX/LENGTH [lifetime {VALID_TIME PREF_TIME | infinite}]
    domain-name TEXT
    dns-server ADDRESS
  interface INT
    ipv6 dhcp server POOL_NAME
    ipv6 nd {other-config-flag | managed-config-flag}
    ipv6 dhcp relay destination ADDRESS

show {ip | ipv6} dhcp binding
clear ip dhcp binding {A.B.C.D | *}
show ip dhcp pool POOL_NAME
show ipv6 dhcp pool
show ip dhcp server statistics
```

NAT Services Configuration

```
configure terminal
  interface F0/0
    ip nat inside
  interface S0/0/0
    ip nat outside
  ip nat inside source static INSIDE_LOCAL INSIDE_GLOBAL
  ip nat pool POOL_NAME FIRST_IP LAST_IP netmask M.M.M.M
  access-list 1 permit A.B.C.D [W.W.W.W]
  ip nat inside source list 1 pool POOL_NAME [overload]
  ip nat inside source list 1 interface S0/0/0 overload
  ip nat inside source static {tcp | udp} INSIDE_LOCAL LOCAL_PORT INSIDE_GLOBAL
    > GLOBAL_PORT [extendable] [stateless]

show ip nat translations
show ip nat statistics
```

WLC WLAN Configuration

1. Create a new WLC Dynamic Interface:

- Name (31 or fewer ASCII characters)
- VLAN ID (Integer value 1-1001, 1007-4094 inclusive)

2. Create a new WLC WLAN:

- Profile Name (32 or fewer ASCII characters)
- SSID (32 or fewer ASCII characters)
- WLAN ID (Integer value 1-512 inclusive)

3. Configure the new WLAN:

- Bind the Dynamic Interface to the WLAN.
- Enable the WLAN on the WLC.
- Enable broadcasting of the WLAN SSID by APs.

4. Secure the new WLAN:

- Set Layer 2 Security to **WPA+WPA2**, enabling the WPA2 checkbox.
- Set WPA2 Encryption to **AES/TKIP/CCMP/GCMP**.
- Enable the PSK checkbox.
- Set the PSK Format to **ASCII** and enter the PSK value.

8.3 Advanced Cisco IOS Configuration

Layer 2 Forwarding

```
configure terminal
  mac address-table aging-time SECS [vlan VLAN_LIST]
  [no] mac address-table learning [vlan VLAN_LIST]
  mac address-table static MAC vlan VLAN_ID {interface INT | drop}
```

Layer 3 Forwarding

```
configure terminal
  interface INT
    ip address A.B.C.D M.M.M.M [secondary]
    ip directed-broadcast
```

```
show ip interface [INT | brief]
debug ip icmp
```

ARP Configuration

```
configure terminal
  arp A.B.C.D MAC arpa
  [no] ip arp proxy disable
  interface INT
    [no] ip proxy-arp
    [no] ip local-proxy-arp
    arp timeout SECS
```

```
show ip interface INT
show [ip] arp
show arp [A.B.C.D detail]
```

```
{clear | debug} arp
```

GRE Tunneling / MTU / MSS Configuration

```
configure terminal
  system mtu [jumbo | routing] BYTES
  ip tcp mss BYTES
  interface L0
    ip address 1.1.1.1 255.255.255.255
  interface F0/0
    mtu BYTES
    {ip | ipv6} mtu BYTES
    no ip {redirects | unreachable}
  interface tunnel0
    ip address 192.168.1.1 255.255.255.252
    tunnel source F0/0
    tunnel destination A.B.C.D
    tunnel path-mpu-discovery
    ip tcp adjust-mss BYTES
  ip route 2.2.2.2 255.255.255.255 192.168.1.2

show interfaces INT
show ip interface INT
show ip traffic interface INT
show system mtu
show tcp
debug ip tcp {adjust-mss | transactions}
```

CEF / dCEF Configuration

```
configure terminal
  [no] {ip | ipv6} cef [distributed]
  ip cef load-sharing algorithm {dpi | include-ports | original | tunnel |
    > universal}
  interface F0/0
    ip load-sharing {per-destination | per-packet}

show processes cpu
show running-config all
show cef [interface INT]
show ip cef [A.B.C.D[/CIDR] | INT] [detail]
clear cef table ipv4
show adjacency [detail]
```

BGP Routing Configuration

```
configure terminal
  router bgp LOCAL_AS
    neighbor A.B.C.D remote-as AS_VALUE
    network A.B.C.D M.M.M.M

show ip bgp
```

Multicast Configuration

```
configure terminal
  {ip | ipv6} multicast-routing
  {ip | ipv6} pim rp-address A.B.C.D
  ip pim send-rp-announce INT scope TTL
```

```
ip pim send-rp-discovery scope TTL
interface F0/0
  ip pim {dense-mode | sparse-mode | sparse-dense-mode}
interface F0/1
  ip igmp join-group A.B.C.D

show ip mroute
show ip pim rp
ping A.B.C.D repeat COUNT
```